The Truncated Icosahedron Star (TI Star)

A doubling cycle and a golden cycle grown from the truncated icosahedron's own pyramids



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The ideas and names are DICTO's. Computation and proof with Claude (Anthropic).

Abstract. The truncated icosahedron (TI) is the midpoint cell of DICTO's path between the White and the Red Queen (#76) and a station of his golden cycles. DICTO proposed filling its core with pentagonal and hexagonal pyramids, and then starting a new golden cycle from TI with those pyramids turned outwards, a spiked core. Cut from its centre, TI is exactly 12 pentagonal and 20 hexagonal pyramids; the hexagonal one carries on every base edge half the icosahedron's dihedral angle, the angle of the Cleo-Icos, because the hexagons of TI lie in the face planes of an icosahedron three times its edge. Turned outwards, the pyramids give the TI Star. **Main theorem.** Put the tip of each spike at s times the centre of its face. For s = 2 (each spike the mirror image of the pyramid inside) and for s = φ (each spike 1/φ of its height), TI, the TI Star, the hull of the 32 tips and TI s times larger form one turn of an endless cycle, each layer filled exactly by tetrahedra of three kinds over the corners and the two kinds of edge: for s = 2 a doubling cycle, the TI Star twice the volume of TI as DICTO's Cleo star is twice the icosahedron's; for s = φ a golden cycle, the TI Star φ times TI and one turn φ^3 in volume. The hull of the tips is a pentakis dodecahedron, not the Catalan one; inside it each turn passes through two pairs of Platonic shells, the dodecahedron and icosahedron of TI's own face planes and those of the spike tips, each pierced by spikes, with a new core between them, the icosahedron truncated at $1/\varphi^3$ of each edge (eleven solids in all, complete in a stated sense; DICTO's growth rule carries each outer pair to the next inner pair). About one sphere the middle of every turn is its own polar, so the polar cycle shares four solids with the cycle and adds three, fourteen per turn. The hull of the tips, which DICTO named the Star Halo, starts a cycle of its own, and spiking each hull in turn gives a star chain that returns to the shape of the truncated icosahedron after four steps: read the other way it is the orbit of TI under DICTO's growth rule of Universal PiTet (#38), a PiTet cycle of its own. Version 2 adds DICTO's idea of giving the core its own pyramids and spikes: its turn gives back the truncated icosahedron, so TI and its core alternate as the cores of a cycle of period two, because each turn swaps the distances of the pentagon and hexagon planes exactly. The spike tips touch none of the twelve stations of DICTO's golden cycle (#54) at any golden scale, so the two cycles are new and separate. All volumes are exact; 333 pieces per turn are free of overlap.

1. Introduction

A truncated icosahedron can be cut from its centre into pyramids, one on every face, as Aristotle's pyramids cut the Platonic solids. Turn the pyramids outwards and it grows spikes. Wrap the spikes and the solid comes back, larger.

The truncated icosahedron appears twice in DICTO's work. In the Spheres of Dharma (#23) and in the twelve-station golden cycle (#54) it is a station between the rectified truncated icosahedron and the Hecatodihedron; in the path between the queens (#76) its four-dimensional counterpart, the bitruncated 600-cell, is built from 120 truncated icosahedra and 600 truncated tetrahedra. Paper #77 showed that, face to face, the five Platonic face pyramids keep to their solids. DICTO proposed the same cut for TI, and then a new golden cycle that begins, like his Cleo star (#35), with the pyramids turned outwards. This paper carries out both.

2. The truncated icosahedron from its centre

With edge 1, the truncated icosahedron TI (corners: the even permutations of (0, ±1, ±3φ), (±1, ±(2 + φ), ±2φ), (±φ, ±2, ±(2φ + 1)), halved) has volume (125 + 43√5)/4. Cut from its centre it falls into the pyramids of Table 1, exactly.

Pyramid	Count	Height	Volume	Base-edge angle
TI, pentagonal	12	√(1250 + 410√5)/20 = 2.32744	1.33477	73.528° ($\cos^2 = (5 + \sqrt{5})/90$)
TI, hexagonal	20	√(42 + 18√5)/4 = 2.26728	3√(14 + 6√5)/8 = 1.96353	69.095° on every edge ($\cos^2 = 1/2 - \sqrt{5}/6$): $\theta_I/2$
truncated tetrahedron, triangular	4	5√6/12 = 1.02062	5√2/48	74.207° ($\cos^2 = 2/27$)
truncated tetrahedron, hexagonal	4	√6/4 = 0.61237	3√2/8	35.264° on every edge: α/2

Table 1. The face pyramids of the two midpoint cells of #76 (edge 1). The base-edge angle is the angle between the base and the plane through the edge and the centre; $\theta_I = \arccos(-\sqrt{5}/3)$ is the icosahedron's dihedral angle and $\alpha = \arccos(1/3)$ the tetrahedron's. Both pyramids of TI have the same side triangle (legs √(58 + 18√5)/4 = 2.47802, the radius of TI).

Proposition 1. The hexagonal pyramid of TI is the Cleo-Icos of edge 3 with its three corners cut off; the truncated tetrahedron's hexagonal pyramid is the tetrahedron's face pyramid of edge 3 with its corners cut off. *Proof.* Truncating a solid keeps the planes of its faces: the hexagons of TI lie in the face planes of the icosahedron of edge 3, and the truncated tetrahedron's in those of the tetrahedron of edge 3. A face pyramid's angle along an edge of its base depends only on the face plane and the edge, so the angles are those of the Cleo-Icos and of the tetrahedron's pyramid (#44, #77). ∎

Base to base, the hexagonal pyramid of TI on the truncated tetrahedron's makes a hexagonal bipyramid of height (√6 + √(42 + 18√5))/4 = 2.87966: the meeting of the two midpoint cells of #76, as the tetrahedron's pyramid meets the octahedron's in the octet. Face to face the four pyramids keep to their solids, as in #77: round a side edge only one pentagonal and two hexagonal pyramids close (111.381° + 2 × 124.309° = 360°; for the truncated tetrahedron 67.115° + 2 × 146.443°), and the angles round an edge exclude TI from every face-to-face tiling by the two solids (the only edge star of up to five solids is 2 × 109.47° + 2 × 70.53° of truncated tetrahedra; the closest one with TI misses by 1.57°).

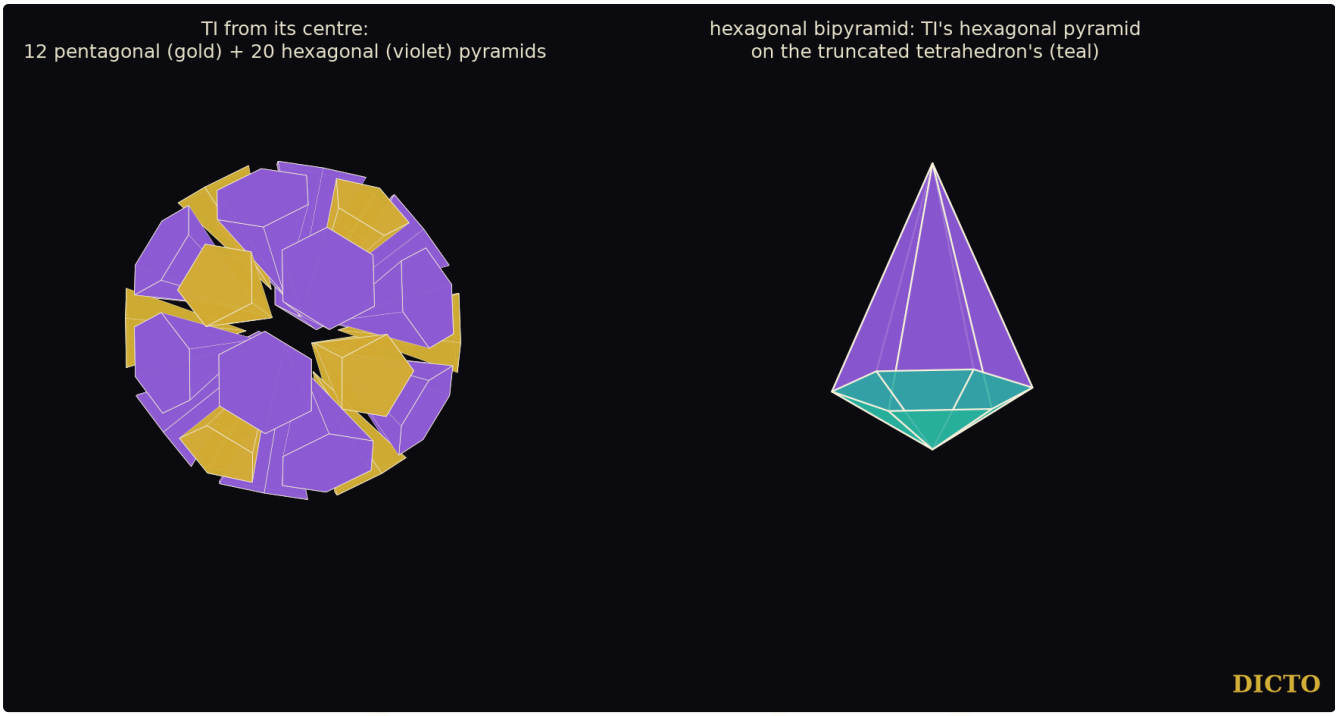


Figure 1. Left: the truncated icosahedron cut from its centre, 12 pentagonal (gold) and 20 hexagonal (violet) pyramids drawn apart. Right: the hexagonal bipyramid, the hexagonal pyramid of TI on the truncated tetrahedron's (teal).

3. Main theorem: the TI Star and its two cycles

Turn every pyramid outwards and the solid becomes a star. The tips of the star mark the faces of a larger one, and the space between them fills with small tetrahedra of only three kinds.

Let c_f be the centre of face f of TI (it is the foot of the perpendicular from the centre, because the faces are regular). For $s > 1$ the **TI Star** TI(s) (the Double TI Star for $s = 2$, the Golden TI Star for $s = \varphi$; DICTO's names, after his Cleo star) is TI with a pyramid over every face whose tip is $s \cdot c_f$; its spikes are the inward pyramids scaled in height by $s - 1$, so TI(s) has volume $s \cdot$TI. Let K(s) be the convex hull of the 32 tips (the Star Halo, DICTO's name, Section 6), and $s \cdot$TI the truncated icosahedron scaled by s; its face centres are exactly the tips.

> **Theorem.** For $s = 2$ and for $s = \varphi$: (a) TI $\subset$ TI(s) $\subset$ K(s) $\subset s \cdot$TI, and K(s) is a pentakis dodecahedron: 32 corners (the tips), 60 congruent triangles, each over one corner of TI. (b) The layer K(s) minus TI(s) is filled exactly by 60 corner tetrahedra, each a corner v of TI with the three tips over the faces at v, and 90 edge tetrahedra, each an edge vw of TI with the two tips over the faces at vw (60 over pentagon–hexagon edges, 30 over hexagon–hexagon edges). (c) The layer $s \cdot$TI minus K(s) is filled exactly by 60 corner tetrahedra (a corner of $s \cdot$TI and the three face centres there) and 90 edge tetrahedra (an edge of $s \cdot$TI and the two face centres beside it). (d) Scaling by s repeats the turn without end, so the four stations form a cycle: for $s = 2$ a doubling cycle, for $s = \varphi$ a golden cycle.

Station or piece	Count	Doubling, s = 2	Golden, s = φ
TI	1	1	1
TI Star TI(s)	1	2	φ = 1.618034
pentakis hull K(s)	1	6.067694	3.212896
s·TI (next turn)	1	8	φ³ = 4.236068
layer 1: corner tetrahedron	60	0.041839	0.014743
layer 1: edge tetrahedron, pentagon–hexagon	60	0.017132	0.007892
layer 1: edge tetrahedron, hexagon–hexagon	30	0.017647	0.007892
layer 2: corner tetrahedron	60	0.017451	0.009240
layer 2: edge tetrahedron, pentagon–hexagon	60	0.008725	0.004620
layer 2: edge tetrahedron, hexagon–hexagon	30	0.012058	0.006385

Table 2. Volumes as multiples of TI. One turn has 1 + 32 + 150 + 150 = 333 pieces (the core, the spikes and the two layers). In the frame of #54 (edge of TI 1/φ), the hull K(2) has volume 79.193496 and the layer-1 pentagon–hexagon tetrahedron 0.2236068, agreeing with 30 + 22√5 and √5/10 to nine digits; for s = φ both layer-1 edge tetrahedra have the same volume, and the layer-2 hexagon–hexagon tetrahedron 0.0833333, 1/12 to nine digits. These closed forms are recognised from the numbers, not proved.

Proof and checks. The tips are s times the face centres, so they are the face centres of s·TI, which gives (a) together with the computed hull: its 60 face planes are all at one distance, each through one pentagon tip and two hexagon tips over one corner of TI. Each corner tetrahedron of layer 1 has its base on a face of K(s) and its apex at the corner of TI below; each edge tetrahedron is bounded by the two spike faces over its edge and by two faces of the neighbouring corner tetrahedra; so the pieces close round every spike edge and every hull edge. The volumes of each layer add up to the layer to 10^{-12} for both s (corners exact in $\mathbb{Q}(\sqrt{5})$; volumes in double precision), and all 333 pieces of a turn, the core and the spikes are pairwise free of overlap by the separating-axis test (faces and edge cross products of every pair of convex pieces whose bounding spheres meet). This is (b) and (c); (d) is the scaling. ∎

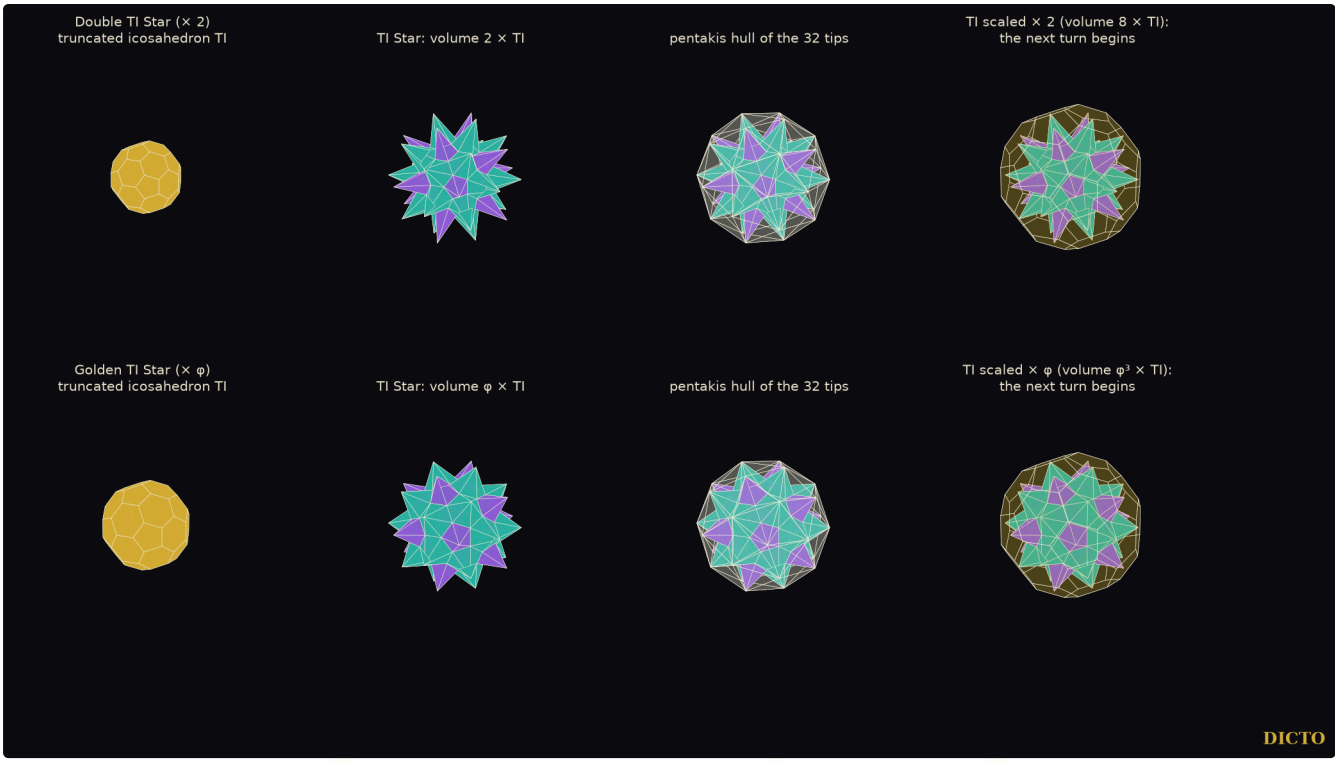


Figure 2. One turn of each cycle, each row to its own scale. Top, the Double TI Star (s = 2); bottom, the Golden TI Star (s = φ). In each row: TI; the TI Star (pentagonal spikes violet, hexagonal teal); the pentakis hull of the 32 tips; and TI scaled by s, whose face centres are the tips, where the next turn begins.

4. All the solids of one turn

Inside each turn two pairs of Platonic shells appear, a dodecahedron and an icosahedron each time, and between them a new truncated icosahedron.

A turn has three sets of points, the 60 corners of TI, the 12 pentagon tips and the 20 hexagon tips, and two sets of planes, the 12 pentagon and the 20 hexagon planes of TI. The 12 pentagon tips are the corners of a regular icosahedron I_p, the 20 hexagon tips those of a regular dodecahedron D_h; the pentagon planes of TI bound a regular dodecahedron D_{TI}, its hexagon planes a regular icosahedron I_{TI} (the icosahedron of edge 3 of Proposition 1).

> **Proposition 2 (the solids of one turn).** For s = 2 and s = φ:
>
> (a) The convex solids of the turn nest as TI = $D_{TI} \cap I_{TI} \subset D_{TI}$, $I_{TI} \subset I_p \cap D_h \subset I_p$, $D_h \subset K \subset s{\cdot}TI = s{\cdot}D_{TI} \cap s{\cdot}I_{TI}$. The two solids of each pair cross each other; TI with its 12 pentagonal spikes lies in I_p, TI with its 20 hexagonal spikes in D_h, and the TI Star in K; all 32 tips lie outside D_{TI} and I_{TI}.
>
> (b) The 12 pentagonal spikes pierce the faces of D_h and the 20 hexagonal spikes those of I_p, so K is at once D_h with 12 pentagonal pyramids and I_p with 20 triangular pyramids: a spiked dodecahedron and a spiked icosahedron.
>
> (c) The core $I_p \cap D_h$ has 12 pentagons and 20 hexagons and all its corners on one sphere; its hexagons have alternating edges in the ratio √5, so it is the icosahedron truncated at $1/\varphi^3$ of each edge (the uniform TI is truncated at 1/3). It has the same shape in both cycles.

(d) DICTO's growth rule of *The Golden Cycle* (each corner of a solid gives a face plane of the next) carries K to s·TI, I_p to s·D_{TI} and D_h to s·I_{TI}: the outer pair of one turn gives the inner pair of the next.

(e) The list is complete in three senses: it holds every convex hull of a union of the three point sets and every intersection of these hulls and of the plane sets; of the 256 solids made of whole symmetric families of the turn's pieces (TI always included), only TI, K and s·TI are convex, and the hulls of all 256 are TI, I_p, D_h, K and s·TI; and one step of the growth rule from any solid of the list gives a solid of the list or of the next turn, except from TI, I_{TI} and D_{TI}.

Solid	Double, s = 2	Golden, s = φ
TI	1	1
TI with its 12 pentagonal spikes	1.289707	1.179049
TI with its 20 hexagonal spikes	1.710293	1.438985
TI Star	2	φ
inner icosahedron I_{TI} (hexagon planes)	1.065440	1.065440
inner dodecahedron D_{TI} (pentagon planes)	1.265673	1.265673
core $I_p \cap D_h$	4.103174	2.172666
outer icosahedron I_p (pentagon tips)	4.626695	2.449874
outer dodecahedron D_h (hexagon tips)	4.697099	2.487154
K, the pentakis hull	6.067694	3.212896
s·TI	8	φ^3

Table 3. The eleven solids of one turn, as multiples of TI. Inclusions were tested for every pair (all corners of one inside every face plane of the other, to 10^{-9}); the edge ratio of the core is √5 to 10^{-8} in both cycles; the growth-rule identities hold to 10^{-9} in volume (study scripts/stations.py, inner_pair.py, families.py, rule_chains.py). The inner pair depends only on TI, so it is the same in both cycles; the outer solids scale with s.

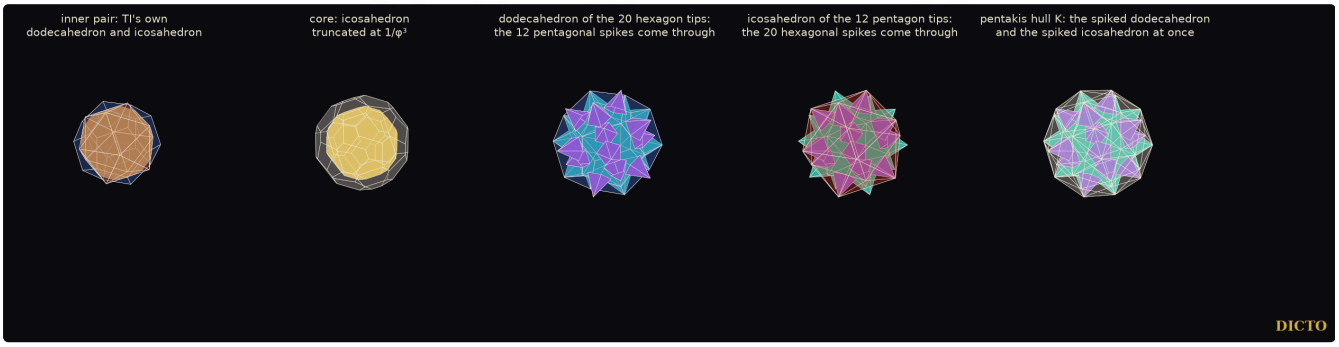


Figure 3. The solids of one turn of the golden cycle. From left: TI (gold) inside its own dodecahedron (blue) and icosahedron (red); the core $I_p \cap D_h$ (translucent) round TI; the outer dodecahedron D_h of the hexagon tips, with the pentagonal spikes coming through its faces; the outer icosahedron I_p of the pentagon tips, with the hexagonal spikes coming through; and K, both at once.

Where the list stops. Repeated, the growth rule leaves the cycle in two side chains that never land on a station: from TI it alternates TI-shaped solids and Catalan pentakis dodecahedra, growing by 1.172168 in length every two steps; from I_{TI} and D_{TI} it alternates icosahedra and dodecahedra, growing by 1.583592 every two steps. Neither factor is a power

of s, so these chains wind through the turns without closing. The turn passes through both Platonic solids of five-fold symmetry twice; the dodecahedra are the shape of the Red Queen's core cell (#65), and whether they can be matched to her nine layers is left open.

5. The polar cycle

Reflect every solid of the turn in a sphere, corners to faces and faces to corners: the middle of the turn comes back as itself.

> **Proposition 3 (the polar cycle).** For s = 2 and s = φ, let $\rho^2 = R(I_p)\cdot r(D_h)$, the circumradius of the outer icosahedron times the inradius of the outer dodecahedron (ρ = 1.337 times the radius of TI for s = φ, 1.653 for s = 2). In the sphere of radius ρ the outer icosahedron and the outer dodecahedron are polar to each other, and so are the Star Halo K (Section 6) and the core: the four middle solids of the turn are their own polar set. The polars of TI, I_{TI} and D_{TI}, scaled by 1/s into the same turn, are a Catalan pentakis dodecahedron and a second icosahedron and dodecahedron, and the two cycles nest together in one space:
>
> $$\mathrm{TI} \subset I_{TI} \subset I^*, \quad \mathrm{TI} \subset D_{TI} \subset D^*, \quad I^*, D^* \subset \text{Catalan pentakis} \subset \text{core} \subset I_p, D_h \subset K \subset s\cdot\mathrm{TI},$$
>
> where I* and D* are the polar icosahedron and dodecahedron (each pair crosses itself, and I_{TI}, D_{TI} cross the polar solid of the other kind where noted in the data): fourteen solids per turn, the four middle ones shared.

The equality $\rho^2 = R(D_h)\cdot r(I_p)$ holds as well, because the icosahedron and the dodecahedron have the same ratio of circumradius to inradius; that is why one sphere serves both. In the polar cycle TI's place is taken by the Catalan pentakis dodecahedron, the polar of the uniform truncated icosahedron. Exact order and every inclusion: study data merged-turn.json; the polar identities hold to 10^{-8} in volume.

6. The Star Halo and the star chain

Treat the hull of the star as a new core, spike it, and wrap it again. After four such steps the truncated icosahedron comes back.

DICTO named K, the hull of the 32 tips, the **Star Halo**. Its 60 triangles all lie at one distance from the centre, so when its own face pyramids are turned outwards (tips at s times the foot of each face's perpendicular) the tips land on a scaled copy of its polar, which is the core shape of Section 4. Repeating the step gives the star chain: each solid's spike tips make the next.

> **Proposition 4 (the star chain).** Let S_0 = TI and S_{n+1} the hull of the spike tips of S_n, the tips at s times the feet of the perpendiculars to its faces. For s = φ and s = 2: S_1 is the Star Halo, S_2 has the shape of the core (the icosahedron truncated at $1/\varphi^3$, edges in the ratio √5), S_3 is the Catalan pentakis dodecahedron, the polar of TI, and S_4 has the shape of TI again, so the chain repeats with period four. Every step is one turn of a cycle of the kind of the Theorem: $S_n \subset \text{star} \subset S_{n+1} \subset s\cdot S_n$, the star of volume s times S_n, both layers filled exactly by pyramids over the corners of S_n and of $s\cdot S_n$ (tetrahedra over three-fold corners, pentagonal and hexagonal pyramids over five- and six-fold ones) and tetrahedra over the

edges; 333 pieces per turn from a truncated icosahedron, 305 from a pentakis dodecahedron.

Relation to DICTO's rule T (#38). Taking the hull of the feet of the faces undoes DICTO's growth rule T(S) = {x : v·x ≤ v·v for every corner v of S} of Universal PiTet (#38): $T(S_{n+1}) = s \cdot S_n$. Read the other way, the star chain is the T-orbit of TI: T carries TI to the Catalan pentakis dodecahedron (a polar step, since TI's corners lie on one sphere), that to the core shape, the core to the Star Halo (again a polar step) and the Star Halo to TI, larger by exactly (17745 − 7712√5)/361 = 1.386271 per turn. This is the structure of DICTO's PiTet Proposition, whose T-orbit of the icosidodecahedron, ID → RT → GTI → FPD → ID, grows by $10/\varphi^4$ per turn: TI has a PiTet cycle of its own.

Step	Solid	Corners · faces	Edge ratio	Hull / S_n, s = φ	Hull / S_n, s = 2
S_0	TI	60 · 12 pentagons + 20 hexagons	1	3.212896	6.067694
S_1	Star Halo	32 · 60 triangles	1.098186	3.376409	6.376497
S_2	core shape	60 · 12 pentagons + 20 hexagons	√5	3.226814	6.093979
S_3	Catalan pentakis dodecahedron	32 · 60 triangles	1.127322	3.452883	6.520921
S_4	TI again	60 · 12 pentagons + 20 hexagons	1	3.212896	6.067694

Table 4. The star chain. Edge ratio: longest over shortest edge. Volumes of each step's hull over the solid it grows from; the star is always s times the solid. Shapes compared up to scale by corner radii and edge lengths; fills exact to 10^{-9} in volume at every step; no overlapping pair among the 333 or 305 pieces of steps 0 to 3 (separating-axis test). After four steps the radius has grown by 4.944274 (s = φ) and 11.541757 (s = 2), that is $s^4 \cdot c$ with c = (17745 + 7712√5)/48505 = 0.7213598 exactly; c is not a power of φ, so the chain returns to the shape of TI but not to a golden or doubled copy of it (study scripts/halo_cycle.py, star_chain.py; data/halo-cycle-*.json, star-chain-*.json).

The scaling numbers. Because every tip is the foot of a perpendicular from the centre to a plane through points of $\mathbb{Q}(\sqrt{5})^3$, the whole chain stays in $\mathbb{Q}(\sqrt{5})$ and its ratios are exact. With the tips at the feet themselves (s = 1; any s multiplies step n by s^n), the squared outer radius changes at the four steps by (445 + 16√5)/545, (37 + 16√5)/89, (645 + 226√5)/1335 and (51 + 18√5)/109, and then by the same four numbers again, step after step (checked exactly to eight steps). Over one period the radius changes by c = (17745 + 7712√5)/48505, with 48505 = 5·89·109; the squared constant is $(7712 + 3549\sqrt{5})^2/(5 \cdot 89^2 \cdot 109^2)$; its inverse, the gain of T over one turn, is (17745 − 7712√5)/361, with $361 = 19^2$. The primes 89 and 109 alternate in the denominators; 109 already appears in TI, whose side triangles have apex angle cos (80 + 9√5)/109 (Table 1 study data). Why 89 appears is left open (study scripts/chain_exact.py, data/chain-exact.json).

Kepler's spikes. A spike rule with no free number is Kepler's: the tip over a face is where the planes of its neighbours first meet its axis. For TI this puts the pentagon tips at the corners of its own icosahedron I_{TI} and the hexagon tips at those of its own dodecahedron D_{TI}, and the hull of the 32 tips is the Catalan pentakis dodecahedron, the polar of TI and its first step under T. That layer fills exactly: TI, its 32 Kepler spikes, 60 corner tetrahedra and 90 edge

tetrahedra make up its polar, with no overlap. The next Kepler step does not fill (on the pentakis the spikes stand outside their own hull); the hulls alone return to the shape of TI after two steps, larger by (105 − 18√5)/55 = 1.177287, so Kepler's rule gives a chain of hulls but no cycle of pieces. If instead each new solid is the hull of the whole star, its corners and its Kepler tips together (DICTO's proposal), every corner and every tip stays on the hull from the second step on: the solid gains a low pyramid on each triangle, the triangles triple (60, 180, 540, 1620) at an unchanged radius, and the gap between star and hull falls from 28 to 4.8, 1.1 and 0.16 per cent of the solid's volume, a refinement towards a rounded limit rather than a cycle (four steps checked; beyond them the floating-point tests are not reliable; study scripts/kepler_full_hull.py). No natural spike rule tried makes the four-step factors exactly 5 or 12; 4.944 and 11.542 remain near values (study scripts/kepler_chain.py, kepler_cycle.py).

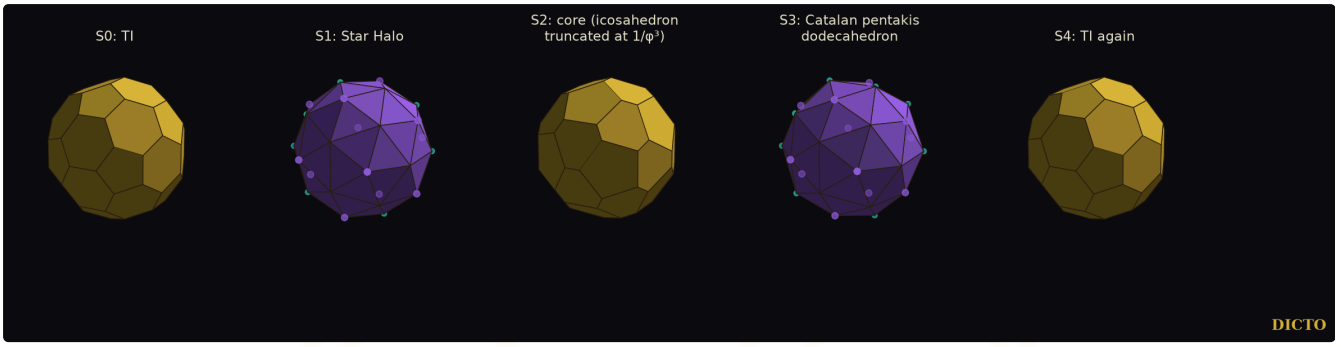


Figure 4. The star chain (golden version), each solid scaled to the same radius: TI; the Star Halo (five-fold corners violet, six-fold teal); the core shape, the icosahedron truncated at 1/φ³; the Catalan pentakis dodecahedron, the polar of TI; and TI again.

The core's own cycle. DICTO proposed giving the core the same treatment as TI: its own pentagonal and hexagonal pyramids from the centre, turned outwards as spikes, and the turn that follows. The core has the faces of TI, 12 pentagons and 20 hexagons, with its planes facing the same directions, so the construction applies unchanged.

> **Proposition 5 (the core's own cycle).** Let C be the core $I_p \cap D_h$ of a turn of TI, and give C its own spikes, the tips at s times the feet of the perpendiculars to its faces, for s = φ and s = 2. Then:
>
> (a) the core of C's turn, the intersection of the icosahedron of its 12 pentagon tips and the dodecahedron of its 20 hexagon tips, is a uniform truncated icosahedron, $s^2(5 + 2\sqrt{5})/15$ times the size of TI in length (5/6 + 11√5/30 = 1.653225 for s = φ, 4/3 + 8√5/15 = 2.525903 for s = 2): taken core to core, TI and C alternate, a cycle of period two;
>
> (b) the reason is a swap of the two plane families. Let q be the square of the ratio of the pentagon distance to the hexagon distance from the centre. Then q(TI) = 13/3 − 22√5/15 = 1.026531^2, q(C) = (195 + 66√5)/361 = 0.974154^2, and q(TI)·q(C) = 1 exactly: each turn puts the new hexagon planes where the pentagon tips were and the new pentagon planes where the hexagon tips were, so the ratio is inverted and two turns restore it; (5 + 2√5)/15 is the square of the ratio of inradius to circumradius of the icosahedron and of the dodecahedron;
>
> (c) the hull of C's 32 tips is the Catalan pentakis dodecahedron, the polar of TI, just as the hull of TI's tips, the Star Halo, is the polar of C; the two halos are partners as the two cores are;

> (d) C's star has s times its volume, and the layer between the star and the hull of its tips fills exactly with 150 pieces of three kinds (a piece over each of the 60 corners to the tips of its three faces, a tetrahedron over each of the 90 edges), with no overlap.

So the cores of the two kinds of turn close in two steps, while the star chain of hulls (Proposition 4) closes in four: TI and its $1/\varphi^3$ core are two faces of one cycle. Parts (a)–(c) were checked exactly in $\mathbb{Q}(\sqrt{5})$ for both s (study 2026-10-30-core-cycle, scripts/core_cycle_exact.py, with the exact arithmetic of the DICTO Solids data set); part (d) to 10^{-9} in volume and by the separating-axis test (scripts/core_cycle.py; data/core-cycle-phi.json, core-cycle-2.json).

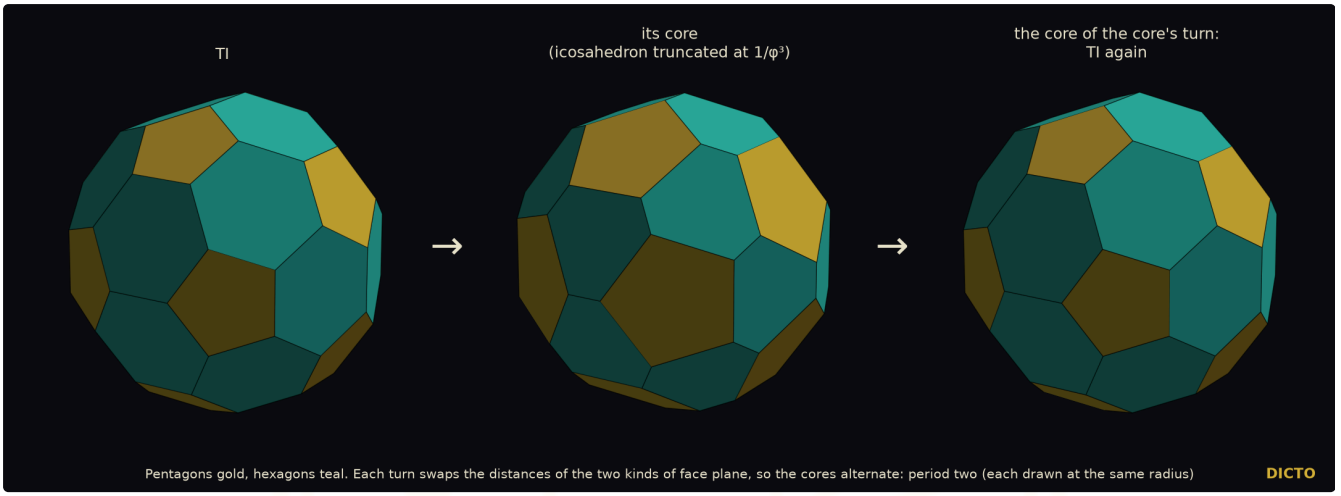


Figure 5. The core's own cycle (golden version), each solid drawn at the same radius, pentagons gold and hexagons teal: TI; its core, the icosahedron truncated at $1/\varphi^3$, whose hexagons have alternating short and long edges; and the core of the core's turn, a uniform TI again. Each turn swaps the distances of the pentagon and hexagon planes from the centre.

7. Doubling and golden, and the Cleo star

Mirroring a face pyramid in its face puts the tip at twice the face centre: that is the doubling cycle, and the TI Star is then exactly twice TI, as DICTO's Cleo star, the icosahedron with its twenty Cleo-Icos turned outwards, is twice the icosahedron (#35, #54). The golden ratio enters when the spikes are $1/\varphi$ of the pyramids' height. The two cycles behave differently from the Cleo star in one respect: the Cleo star's twenty tips are the corners of a dodecahedron, D′, which is how it joins DICTO's twelve-station cycle; the 32 tips of the TI Star lie on two spheres (the pentagon tips 1.0265 times further out than the hexagon tips, the inverse of the Catalan pentakis dodecahedron's proportion), and they touch no station of the twelve-station cycle at any scale φ^k, $k = -2, \ldots, 4$ (study data contacts.json). The TI Star therefore starts a cycle of its own, beside the golden cycle rather than on it.

8. What is new and what is known

- **Known, credited:** DICTO's growth rule T and the PiTet Proposition (#38), of which the star chain is the case of TI; the Catalan pentakis dodecahedron; cutting a convex solid from its centre into face pyramids; raising pyramids on the faces (Kepler's stellations, the Kleetope construction), and the pentakis dodecahedron as the Kleetope of the dodecahedron; the truncated icosahedron and truncated tetrahedron (Archimedes, Kepler); the bitruncated 600-cell of #76 (Boole Stott 1910).

- **New, as far as the sources checked show:** Proposition 1 (the two hexagonal pyramids carry the Cleo-Icos's and the tetrahedron pyramid's angles), the hexagonal bipyramid of the two midpoint cells, the exclusion of the pyramids of TI from face-to-face tilings, and the Theorem: the doubling and golden cycles of the TI Star with their exact fills; Proposition 2, the eleven solids of one turn, the $1/\varphi^3$ core and the growth-rule links; Proposition 3, the self-polar middle and the merged turn of fourteen solids; Proposition 4, the Star Halo cycle and the star chain of period four, the T-orbit of TI with its exact gain; Proposition 5 (Version 2, DICTO's idea), the core's own cycle: TI and its $1/\varphi^3$ core alternate as cores, with period two.
- **Not decided:** which other factors s give a cycle with the same pieces (only s = 2 and s = φ were checked); whether the golden cycle can be joined to the twelve-station cycle by an intermediate solid; the meaning of the primes 89 and 109 in the chain's ratios; and the star chains of DICTO's other solids, which a census will take up.

9. Conclusion

DICTO's idea holds both ways. The core of TI fills exactly with pentagonal and hexagonal pyramids, and the hexagonal ones carry the Cleo-Icos's angle. Turned outwards they make a spiked core from which two new cycles grow, a doubling one and a golden one, each repeating without end with one core, 32 spikes and 300 tetrahedra of six kinds per turn. Inside every turn the two Platonic solids of five-fold symmetry appear twice, as an inner and an outer pair, with a new core between them, the icosahedron truncated at $1/\varphi^3$. The middle of the turn is its own polar, and spiking each hull in turn closes into a star chain of four shapes: the truncated icosahedron, the Star Halo, the core and the Catalan pentakis dodecahedron, the orbit of TI under DICTO's own growth rule of #38. Given its own pyramids and spikes, the core turns back into the truncated icosahedron, so the two alternate as the cores of a cycle of period two.

> **Sources checked** (30 October 2026). *Reference library* (ask-library): "truncated icosahedron face pyramids decomposition from the centre space filling truncated tetrahedron" (returned Richter-Gebert 2026, "An almost trivial observation about the icosahedron"; Wikipedia, "Rhombic dodecahedron"; Sommerville 1922; Messer 1995; Kepler 1619 p. 271; Al-Siyabi, Koca and Koca 2020 p. 6, two copies; Goldberg 1937 p. 1; none treats the face pyramids of TI or cycles through them). DICTO's papers #23, #35, #44, #54, #76, #77, cited for their results. The Kleetope construction and the pentakis dodecahedron are cited for their well-known content, not re-read.

> **Data and code.** Study 2026-10-30-ti-pyramids: scripts/pyramids.py (exact pyramids, angles, side-edge wedges, edge stars), scripts/figure.py, data/pyramids.json. Study 2026-10-30-spiked-ti: scripts/spiked.py (spikes, hull, stations), scripts/contacts.py (contacts with the twelve stations at golden scales), scripts/layers.py (SCALE=2 or SCALE=phi: both layers, volumes, overlaps), scripts/stations.py, inner_pair.py, families.py (all 256 unions of whole families), rule_chains.py (the eleven solids, their inclusions and the growth rule), scripts/figure_both.py, scripts/figure_turn.py; data/layers-2.json, data/layers-phi.json, data/stations-turn.json, inner-pair.json, families-2.json, families-phi.json, rule-chains.json, polar.json, polar-sphere.json, merged-turn.json, halo-cycle-phi-phi.json, halo-cycle-2-2.json, star-chain-phi.json, star-chain-2.json, chain-exact.json, kepler-chain.json, kepler-cycle.json, kepler-full-hull.json, data/contacts.json, data/stations-exact.json (from study 2026-10-25-gnomon-shapes).